

MATHCOUNTS®

2025-2026 School Handbook

RESOURCES + INFO EVERY COACH NEEDS



Look through this handbook and your School Competition Kit.



Log in to your ***Coach Dashboard*** (instructions emailed to you).



Hold weekly, biweekly or monthly practices with your Mathletes.
Use ***Coach Dashboard*** resources to:

Recruit Students

- How to promote competition math in your school
- How to support competitors of different backgrounds and skill levels

Run Meetings

- What to do during a math team practice
- Suggested calendars
- Tips for making practices fun and easy to plan

Get Math Resources

- Problem of the Week
- Pre-written Practice Plans
- Mini video lessons
- How to use this handbook during team practices



Hold the 2026 School Competition (PDF available Nov. 6, 2025 on the ***Coach Dashboard***)



Get ready for the Chapter Competition! On your ***Coach Dashboard***, find:

- Past School + Chapter competitions
- Chapter coordinator contact info
- How to add competitor info for the chapter event
- What to expect on Competition Day



Take your Mathletes to the 2026 Chapter Competition!



**LOG IN TO THE
COACH DASHBOARD**





GET THE HANDBOOK!

Registered 2025-26 Coaches (250 problems + solutions):

download at mathcounts.org/dashboard

Anyone (200 problems + solutions): purchase at mathcounts.org/store

A HANDBOOK PREVIEW

The following three pages provide a preview of the problems in the complete School Handbook. Answers are at the bottom of this page.

Refer to the last page of this preview for an explanation of all resources (including step-by-step solutions) in the complete School Handbook.

STRETCHES

Each covers a particular math topic.



Prepare students for all rounds: Sprint, Target, Team & Countdown.

WARM-UPS

10 problems
no calculators



Prepare students for the Sprint Round.

WORKOUTS

10 problems
calculators encouraged



Prepare students for the Target Round.

Throughout this year's handbook, look for special Countdown Round problems top Mathletes tackled in last year's Chapter  and State  competitions.

CHANCE TO WIN STRETCH

1. 25
2. $\frac{2}{9}$
3. 0.42
4. 7
5. $\frac{1}{6}$
6. $\frac{2}{3}$
7. $\frac{3}{13}$
8. 0.53
9. 57
10. $\frac{5}{9}$

WARM-UP 1

31. 38,000
32. 8
33. G
34. 58
35. 80
36. 6
37. $\frac{2}{3}$
38. 24
39. $\frac{1}{4}$
40. 5

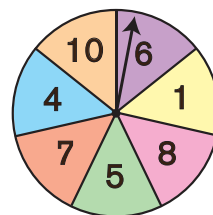
WORKOUT 1

141. 375
142. 1448
143. 4
144. 240
145. 7881
146. $\frac{12}{25}$
147. 1980
148. 754
149. 3
150. 97



Chance to Win Stretch

1. _____ % Kima flips a fair nickel twice and will win \$2 only if both flips come up tails. What is the probability that Kima will win \$2? Express your answer as a percent.
2. _____ Raja has a fair standard six-sided die with four sides labeled “4” and two sides labeled “2.” He gives the die to Ravi and tells him to roll it twice. Raja tells him if he rolls a 4 followed by a 2, he will win \$42. What is Ravi’s chance of winning the \$42? Express your answer as a common fraction.
3. _____ Rosalie, Sara and Trent each flip three fair coins. A player wins a free ticket to the zoo if all three of their coin flips match (all heads or all tails). What is the probability that exactly one of them wins a ticket? Express your answer as a decimal to the nearest hundredth.
4. _____ A bowl contains two red, two green and two purple candies. Christopher randomly selects two candies without replacement. If he does not pick both a red and a green candy, he wins two extra candies. If the odds of winning to not winning is given as ratio $a:b$ in simplest form, what is the value of $a - b$?
5. _____ Blake has three cards—one labeled Y, one labeled T and one labeled O. He shuffles them and places them face down. He then flips them over one at a time, from left to right. If the cards appear in the order T-O-Y, he wins a toy from The Toy Temple. What is the probability that Blake wins? Express your answer as a common fraction.
6. _____ Mark and Mary each roll a fair standard six-sided die simultaneously. The player with the higher number wins the round. If they roll the same number, they repeat the process of simultaneous rolls until there is a winner. What is the probability that Mary wins given that Mark rolls a 2 on the first roll? Express your answer as a common fraction.
7. _____ A randomly selected card from a shuffled standard 52-card deck lands face down on the floor. If Granger correctly guesses that it is a face card (Jack, Queen and King), he wins two new decks of cards. What is the probability that Granger wins? Express your answer as a common fraction.
8. _____ Karina randomly selects two cards, without replacement, from a standard 52-card deck. She wins free tickets to her two favorite rides at the county fair if the cards are either the same color or the same rank (2–10, J, Q, K, A). What is the probability that Karina wins? Express your answer as a decimal to the nearest hundredth.
9. _____ % The spinner shown is divided into seven equal sections labeled with numbers. A player calls out “even” or “odd” before spinning, and if the spinner lands on a number matching the call, the player wins. Jenna calls out “even” and spins. What is the probability that she wins? Express your answer to the nearest percent.
10. _____ A fair standard six-sided die has each side painted a different color. Sara rolls the die three times, and she wins two movie tickets if all three rolls show different colors. What is the probability that Sara wins? Express your answer as a common fraction.





Warm-Up 1

31. _____ Elmina wants to estimate the sum $5394 + 32,147 + 792$ to check if her final answer is reasonable. She rounds each number to the nearest thousand before adding them. What is her estimated sum?
32. _____ An arithmetic sequence has a common difference between consecutive terms. For example, the sequence 5, 12, 19, 26, 33 is an arithmetic sequence with a common difference of 7. The sum of the first six terms in a particular arithmetic sequence is 42. If the first term is 2, what is the fourth term?
33. _____ Mr. Huang's class creates a banner to hang around the walls of his classroom. The banner repeats the 26 letters of the alphabet. What is the 137th letter written on the banner?

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z A B C

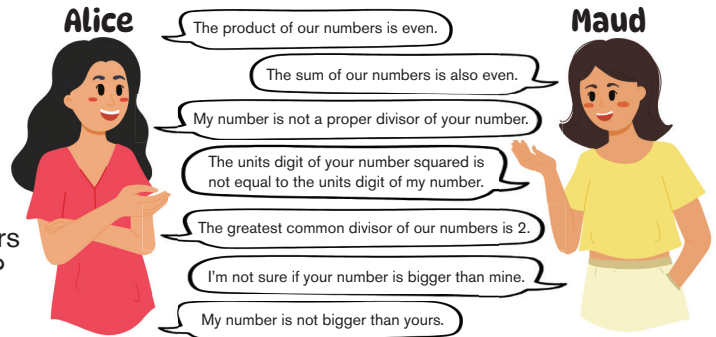
34. _____ If $x + 3 = 10$, what is $x^2 + 3^2$?

35. _____ fluid
 ounces



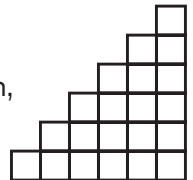
Nadine started with 1 gallon of water. She removed 2 quarts, then added 3 pints and finally removed 4 cups. How many fluid ounces of water are left?

36. _____ Friends Alice and Maud each think of a different integer from 1 to 10, inclusive. They have the following conversation in front of Nina, who confirms to them that the two integers are different. What is Alice's integer?



37. _____ cups Alex is making soup. The recipe's final step, after blending ingredients into a puree, says: "For every $\frac{3}{4}$ cup of puree, add $\frac{1}{3}$ cup of milk and $\frac{1}{2}$ cup of water." If Alex has 3 cups of puree, how many more cups of water than milk does he need to add? Express your answer as a common fraction.

38. _____ units Stephen has 21 unit square tiles, which he aligns in a triangular formation, shown right. What is the perimeter of his figure, in units?



- ★ 39. _____ If Dante cut his blueberry pie into 6 equal slices, and Joe ate $1\frac{1}{2}$ pieces, what fraction of the pie did Joe eat? Express your answer as a common fraction.

- ★ 40. _____ What is the units digit of the sum $1^2 + 2^2 + 3^2 + 4^2 + 5^2$?



Workout 1

141. _____ calories A certain brand of almond milk has 30 calories per cup. If there are 8 fluid ounces in a cup, and the carton holds 100 fluid ounces, how many calories are in a full carton of almond milk?

142. _____ digits What is the total number of digits used when the first 500 positive even integers are written?

143. _____ in² Two photographs have equal areas of 16 square inches—one is a square measuring 4 inches by 4 inches, and the other is a rectangle measuring 2 inches by 8 inches. Each photograph is surrounded by a wooden frame that is 1 inch wide on all sides. What is the absolute difference between the areas of the two frames?

144. _____ miles A car can travel from Sprintville to Countdown City in 3 hours, traveling at a constant rate. If it travels at another constant rate that is 10 mi/h faster on the return trip, it can travel back from Countdown City to Sprintville in 20 fewer minutes. How many miles are between Sprintville and Countdown City?



145. _____ Three distinct prime numbers form an arithmetic sequence. If their sum is 111, what is the least possible value of their product?

146. _____ A number is chosen at random from the set of consecutive integers $\{1, 2, 3, \dots, 25\}$. What is the probability that the number chosen is a divisor of 5!? Express your answer as a common fraction.

147. _____ What is the sum of all positive palindromes less than 2025 that are divisible by 99?

148. _____ inches The average height of a professional basketball player is 78 inches, and there are typically 15 players per team. The average height of a professional baseball player is 74 inches, and there are typically 26 players per team. What is the expected absolute difference, in inches, between the combined heights of the players on a professional basketball team and the combined heights of the players on a professional baseball team?



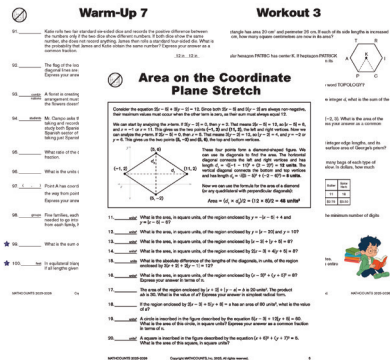
★ 149. _____ integers How many positive integers are divisors of 18, but not of 63?

★ 150. _____ The first three terms of a sequence are 4, 7 and 13. Each term after the first is one less than twice the previous term. What is the sixth term in the sequence?

IN THIS YEAR'S HANDBOOK

Problems

**200 MATH PROBLEMS TO
BOOST PROBLEM-SOLVING SKILLS**



11 WARM-UPS

10 questions per Warm-Up
no calculators used



6 WORKOUTS

10 questions per Workout
calculators used

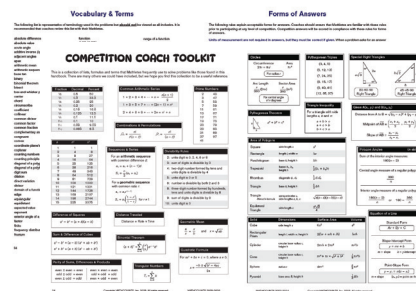


3 STRETCHES

of questions and calculator
use of vary by Stretch

Toolkit

**INCLUDING VOCABULARY
+ FORMS OF ANSWERS**



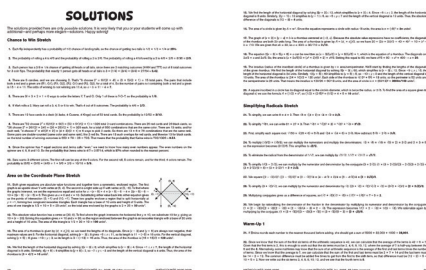
Answers

INCLUDING DIFFICULTY RATINGS

ANSWERS									
Problem #	Answer	Problem #	Answer	Problem #	Answer	Problem #	Answer	Problem #	Answer
1	10	11	10	21	10	31	10	41	10
2	10	12	10	22	10	32	10	42	10
3	10	13	10	23	10	33	10	43	10
4	10	14	10	24	10	34	10	44	10
5	10	15	10	25	10	35	10	45	10
6	10	16	10	26	10	36	10	46	10
7	10	17	10	27	10	37	10	47	10
8	10	18	10	28	10	38	10	48	10
9	10	19	10	29	10	39	10	49	10
10	10	20	10	30	10	40	10	50	10

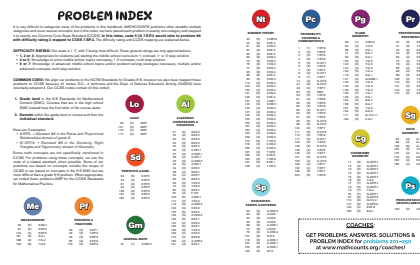
Solutions

200 STEP-BY-STEP EXPLANATIONS



Problem Index + CCSS Mapping

**ALL 200 PROBLEMS CATEGORIZED
+ MAPPED TO THE CCSS**



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Download (if already a registered coach) at mathcounts.org/dashboard

Purchase the handbook at mathcounts.org/store